

GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

SESSION : 2026-27

PRE-BOARD EXAMINATION – I | MARKING SCHEME

CLASS : XII

MAXIMUM MARKS : 50

SUBJECT : COMPUTER SCIENCE (083)

TIME ALLOWED : 1 Hr. 45 Min.

SECTION - A (1 × 13 = 13).

- (a) Gateway (b) FTP (File Transfer Protocol) (c) Star topology (d) Optical fibre cable
(e) Hertz (Hz) (f) pickle (g) writerows() (h) 'r' (read mode, text) (i) `F.seek(20)`
(j) 9 8 (k) None (l) i. Both A and R are True and R is the correct explanation for A.
(m) i. Both A and R are True and R is the correct explanation for A.

SECTION - B

- Q.2** (½ mark each) (a) Simple Mail Transfer Protocol (b) Uniform Resource Locator (c) (2)
eXtensible Markup Language (d) Transmission Control Protocol / Internet Protocol
- Q.3** [11, 10, 3] 3 6 (2)
(1 mark for the list, ½ each for 3 and 6. The list is changed in place because it is mutable;
val is unchanged because the integer is immutable.)
- Q.4** `write()` writes a single string to the file, e.g. `f.write("Hello\n")`. (2)
`writelines()` writes all the strings in a list or other iterable, e.g.
`f.writelines(["A\n", "B\n"])`.
(1 mark for the difference, ½ mark for each example)

SECTION - C

- Q.5** (a) A hub broadcasts incoming data to all ports. A switch sends data only to the intended (3)
destination port, using MAC addresses.
(b) A web browser is client software that requests and displays web pages, e.g. Chrome. A
web server is software or a computer that stores websites and sends pages in response to
requests, e.g. Apache.
(c) HTML is used to display data and has predefined tags. XML is used to store and
transport data and has user-defined tags.
(1 mark each; any one valid point)
- Q.6** (a) Twinkle (1) (3)
(b) 4 Up (1)
(c) `twinkle` then 15 on the next line (½ + ½)

Q.7 **(3)**

```
def FREQ(L):
    D = {}
    for i in L:
        if i in D:
            D[i] += 1
        else:
            D[i] = 1
    return D
```

```
print(FREQ([4, 2, 4, 7, 2, 4]))
```

(Function header ½, loop 1, dictionary update 1, return ½. Accept `D[i] = D.get(i, 0) + 1` or any other correct logic.)

SECTION - D

Q.8 **(4)**

```
import pickle

def addrec():
    f = open("STUDENT.DAT", "ab")
    adm = int(input("Admission No: "))
    name = input("Name: ")
    per = float(input("Percent: "))
    pickle.dump([adm, name, per], f)
    f.close()

def countrec():
    count = 0
    f = open("STUDENT.DAT", "rb")
    try:
        while True:
            rec = pickle.load(f)
            if rec[2] > 75:
                print(rec)
                count += 1
    except EOFError:
        f.close()
    return count
```

(a) 2 marks: open in 'ab' ½, input ½, dump 1. (b) 2 marks: open in 'rb' and loop ½, load with EOFError ½, condition and display ½, count and return ½.

Q.9 **(4)**

(a) ARPANET (Advanced Research Projects Agency Network) was the first packet-switched network, started in 1969 by the US Department of Defense. NSFNET evolved from it. (1)

(b) Circuit switching sets up a dedicated physical path before transmission, which stays reserved for the whole session. Packet switching has no dedicated path: data is split into packets that may travel by different routes. (1)

(c) Microwaves travel in straight lines and cannot bend around obstacles, so the sending and receiving antennas must be aligned and in direct sight of each other. (1)

(d) An IP address is a unique numeric address assigned to each device on a network to identify it, e.g. 192.168.1.10. (1)

Q.10 (a)

(4)

10 15
25 40

(1 mark per line)

(b)

Riya-85-A
Amit-40-B
Joy-90-A
Sam-60-B

(½ mark per line)

SECTION - E

Q.11 (a) Module: csv. Function: csv.writer(). (½ + ½)

(5)

```
import csv
def AddBook():
    f = open("LIBRARY.CSV", "a", newline='')
    w = csv.writer(f)
    bid = input("Book ID: ")
    title = input("Title: ")
    author = input("Author: ")
    qty = int(input("Qty: "))
    w.writerow([bid, title, author, qty])
    f.close()

def ShowLowStock():
    count = 0
    f = open("LIBRARY.CSV", "r")
    r = csv.reader(f)
    for rec in r:
        if int(rec[3]) < 5:
            print(rec)
            count += 1
    f.close()
    print("Total books with low stock:", count)
```

(b) 2 marks (c) 2 marks. Also accept a header row skipped with next(r).

Q.12 (a) TECH block. It has the most computers (120), so placing the server there keeps most traffic local (80–20 rule). (1)

(5)

(b) Star layout centred on TECH: TECH–ADMIN (60 m), TECH–HR (50 m), TECH–ACCOUNTS (110 m).

Also accept the minimum-cable layout TECH–HR (50 m), TECH–ADMIN (60 m), HR–ACCOUNTS (70 m). (1)

(c) In the star layout, place a repeater between TECH and ACCOUNTS, because that distance (110 m) is more than 100 m and the signal weakens over that length. In the minimum-cable layout no segment is over 100 m, so no repeater is needed. Give credit for either answer if it matches the student's own layout. (1)

(d) WAN (Wide Area Network). (1)

(e) VoIP (Voice over Internet Protocol). (1)

***** End of Marking Scheme *****